

The Ising Model in One and Two Dimensions

Exact results and Monte Carlo simulations

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Abstract

The Ising model is the simplest model of a system in which many interacting particles collectively undergo a phase transition. These notes give a self-contained introduction to the model in one and two dimensions. After a reminder of the statistical mechanics needed, we solve the one-dimensional chain exactly with the transfer matrix and show that it has no phase transition. We then introduce mean-field theory, and turn to the two-dimensional square lattice: the Peierls argument, Kramers–Wannier duality, Onsager’s exact solution, critical exponents and universality. The second half of the notes is devoted to Monte Carlo simulations: importance sampling, the Metropolis, heat-bath and Wolff algorithms, the statistical analysis of the data, and finite-size scaling. Every numerical result is produced by the accompanying Python code, and checked against the exact solutions. The notes are aimed at undergraduate students who have followed a first course in statistical physics. Exercises with answers are given at the end.

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1 Introduction

1.1 Ferromagnetism

A piece of iron can be a permanent magnet: its atomic magnetic moments, carried by the spins of electrons, point on average in the same direction, even without any external field. This *spontaneous magnetization* disappears when the iron is heated above its *Curie temperature*, 1043 K. Above this temperature the moments point in random directions, and the magnetization vanishes. The transition between the two behaviours is sharp: the spontaneous magnetization goes continuously to zero at the Curie temperature, but its derivative, the magnetic susceptibility and the specific heat behave singularly there.

The alignment is caused by the *exchange interaction*, a quantum-mechanical effect that makes neighbouring spins prefer to be parallel. Thermal agitation opposes this order. The phase transition results from the competition between energy, which favours order, and entropy, which favours disorder. The Ising model is the simplest model that captures this competition.

1.2 The model

Consider a lattice of N sites. On each site i sits a “spin” s_i that can only take two values, $s_i = +1$ (up) or $s_i = -1$ (down). A *configuration* of the system is the list $\sigma = (s_1, \dots, s_N)$; there are 2^N of them. The energy of a configuration is

$$H(\sigma) = -J \sum_{\langle ij \rangle} s_i s_j - h \sum_{i=1}^N s_i, \quad (1)$$

where $\sum_{\langle ij \rangle}$ runs over all pairs of nearest neighbours, each pair counted once.

- The coupling J measures the interaction between neighbours. For $J > 0$ the energy is lowest when neighbouring spins are parallel: the model describes a *ferromagnet*. For $J < 0$ it describes an *antiferromagnet*. We take $J > 0$.
- The field h measures the coupling to an external magnetic field. It is the product of the magnetic moment of a spin and the magnetic field. A positive h favours $s_i = +1$.

We study the chain ($d = 1$, each spin has $q = 2$ neighbours) and the square lattice ($d = 2$, $q = 4$), with periodic boundary conditions (except for the open chain of Section 3.1). We measure energies in units of J and temperatures in units of J/k_B , that is, we set $J = k_B = 1$ in all numerical results.

The Ising model is a drastic simplification of a real magnet: the spins are classical, they can point in only two directions, and they only interact with their nearest neighbours. It is nonetheless far from trivial, and, as we shall see, its behaviour near the transition is *exactly* the same as that of many real systems.

1.3 A short history

The model was proposed in 1920 by Wilhelm Lenz [1], and solved in one dimension by his student Ernst Ising in his 1924 thesis [2]. Ising found that the chain has no phase transition, and wrongly suggested that the same was true in any dimension. In 1936, Rudolf Peierls showed that the two-dimensional model does have spontaneous magnetization at low temperature [3]. In 1941, Hendrik Kramers and Gregory Wannier located the critical temperature exactly, using a symmetry of the model called duality [4]. In 1944, Lars Onsager obtained the exact free energy of the two-dimensional model [5], a landmark of theoretical physics. Bruria Kaufman simplified the solution and extended it to finite lattices [6], and Chen Ning Yang computed the spontaneous

magnetization [7]. The three-dimensional model has never been solved exactly. For the history of the model, see [9].

At the same time, the Ising model became the test bench of computer simulations. The Metropolis algorithm, invented in 1953 [10], is still the starting point of most Monte Carlo methods, and the cluster algorithms of Swendsen, Wang [11] and Wolff [12] were first developed and tested on the Ising model and its generalizations.

1.4 Why the Ising model matters

The importance of the model goes far beyond magnetism:

- **Lattice gas.** Writing $s_i = 2n_i - 1$, where $n_i = 1$ if site i is occupied by an atom and 0 if it is empty, maps the Ising model onto a model of a gas of atoms that attract each other [8]. The ferromagnetic transition becomes the liquid–gas critical point.
- **Binary alloys.** With $s_i = \pm 1$ for two kinds of atoms, the model describes the unmixing of alloys ($J > 0$) and, with an antiferromagnetic coupling ($J < 0$), the order–disorder transitions of alloys such as β -brass.
- **Universality.** Near a critical point, the details of the interactions do not matter. The three-dimensional Ising model has *exactly* the same critical exponents as the liquid–gas critical point of carbon dioxide or water, or as uniaxial magnets (Section 5.4).
- **Beyond physics.** Ising-like models describe neural networks (the Hopfield model [17] and Boltzmann machines), image restoration, and opinion dynamics in social systems.

2 Statistical mechanics of the Ising model

2.1 The canonical ensemble

A system in contact with a heat bath at temperature T is found in configuration σ with the Boltzmann probability

$$P(\sigma) = \frac{e^{-\beta H(\sigma)}}{Z}, \quad Z = \sum_{\sigma} e^{-\beta H(\sigma)}, \quad \beta = \frac{1}{k_B T}. \quad (2)$$

The *partition function* Z contains all the thermodynamics, through the free energy $F = -k_B T \ln Z$. The average of any observable A is $\langle A \rangle = \sum_{\sigma} A(\sigma) P(\sigma)$.

2.2 Thermodynamic quantities and fluctuations

The quantities we will compute are the energy and the magnetization,

$$E = \langle H \rangle, \quad M = \langle \sum_i s_i \rangle,$$

and their responses: the specific heat $C = \partial E / \partial T$ and the magnetic susceptibility $\chi = \partial M / \partial h$. They are all derivatives of $\ln Z$:

$$E = -\frac{\partial \ln Z}{\partial \beta}, \quad C = \frac{\partial E}{\partial T} = \frac{\langle H^2 \rangle - \langle H \rangle^2}{k_B T^2}, \quad (3)$$

$$M = \frac{1}{\beta} \frac{\partial \ln Z}{\partial h}, \quad \chi = \frac{\partial M}{\partial h} = \frac{\langle \mathcal{M}^2 \rangle - \langle \mathcal{M} \rangle^2}{k_B T}, \quad (4)$$

where $\mathcal{M} = \sum_i s_i$. These *fluctuation–dissipation relations* are very useful in simulations: they give the responses C and χ from the fluctuations of H and $\mathcal{M}$ at fixed temperature and field,

without having to take numerical derivatives. Since E , M , C and χ are proportional to N , we give their values per spin: $e = E/N$, $m = M/N$, $c = C/N$ and χ/N , which we still call χ .

The difficulty is the number of terms in Z . A 10×10 lattice already has $2^{100} \approx 1.3 \times 10^{30}$ configurations: summing over all of them, even at 10^9 configurations per second, would take 4×10^{13} years. Exact results require mathematical tricks (Sections 3 and 5); otherwise we must sample the configurations (Section 6).

2.3 Symmetry and spontaneous symmetry breaking

At $h = 0$, the energy (1) does not change when all the spins are flipped, $s_i \rightarrow -s_i$. For every configuration with magnetization $\mathcal{M}$ there is one with $-\mathcal{M}$ and the same probability, so

$$\langle \mathcal{M} \rangle = 0 \quad \text{for any finite system at } h = 0.$$

A finite system therefore never has a spontaneous magnetization in the strict sense. A phase transition can only occur in the *thermodynamic limit* $N \rightarrow \infty$. There, the spontaneous magnetization is defined by taking the limit of a small field in the right order,

$$m_0(T) = \lim_{h \rightarrow 0^+} \lim_{N \rightarrow \infty} \frac{M}{N}. \quad (5)$$

If $m_0 \neq 0$, the system has chosen one of the two directions although the energy has no preferred one: the symmetry is *spontaneously broken*. The magnetization is the *order parameter* of the transition. Mathematically, a phase transition is a point where the free energy per spin, $f = F/N$ in the limit $N \rightarrow \infty$, is not an analytic function of T and h . Since each finite Z is a finite sum of exponentials, the non-analyticity can only appear in the limit.

In a large but finite system below the transition, the magnetization does not vanish at a given moment: it fluctuates around $+m_0$ or $-m_0$, and very rarely jumps from one to the other. This is why simulations measure $\langle |m| \rangle$ rather than $\langle m \rangle$. For the same reason, the 2D simulations at $h = 0$ measure the fluctuations of $|m|$,

$$\chi' = \frac{N \left(\langle m^2 \rangle - \langle |m| \rangle^2 \right)}{k_B T}, \quad (6)$$

instead of Eq. (4), which in a finite system at $h = 0$ reduces to $N \langle m^2 \rangle / k_B T$ and grows like N below T_c because m flips between $\pm m_0$. Below T_c , χ' measures the response within one ordered state; above T_c , where m has Gaussian fluctuations, $\chi' = (1 - 2/\pi)\chi$. Both diverge at T_c with the same exponent. All the 2D results at $h = 0$ below use χ' ; the 1D results, and all results in a field, use Eq. (4).

2.4 Warm-up: independent spins

Without interactions ($J = 0$), the spins are independent:

$$Z = \prod_{i=1}^N \sum_{s_i=\pm 1} e^{\beta h s_i} = (2 \cosh \beta h)^N, \quad m = \tanh(\beta h), \quad \chi|_{h=0} = \frac{1}{k_B T}.$$

This is a *paramagnet*: the field aligns the spins against thermal agitation, and the susceptibility follows Curie's law $\chi \propto 1/T$. There is no spontaneous magnetization. Interactions are needed for order.

3 The Ising chain: an exact solution

3.1 The open chain at zero field

Consider first an open chain of N spins at $h = 0$, $H = -J \sum_{i=1}^{N-1} s_i s_{i+1}$. Introduce the *bond variables* $\tau_i = s_i s_{i+1} = \pm 1$, which say whether bond i is satisfied (+1) or broken (-1). A configuration is fully described by s_1 and $\tau_1, \dots, \tau_{N-1}$, and these variables are independent:

$$Z = \sum_{s_1} \prod_{i=1}^{N-1} \sum_{\tau_i=\pm 1} e^{K\tau_i} = 2 (2 \cosh K)^{N-1}, \quad K = \beta J.$$

For large N the free energy, energy and specific heat per spin are

$$f = -k_B T \ln(2 \cosh K), \quad e = -J \tanh K, \quad c = k_B \frac{K^2}{\cosh^2 K}. \quad (7)$$

These are smooth functions of T for all $T > 0$: there is no phase transition. The specific heat has a broad maximum, $c = 0.439 k_B$ at $T = 0.834 J/k_B$, which signals that the spins order gradually as T decreases (Figure 5).

3.2 The transfer matrix

With a field, the bond variables are no longer independent, and we use a more powerful method. For a periodic chain ($s_{N+1} = s_1$) we write the Boltzmann weight as a product of factors that each involve two neighbouring spins:

$$e^{-\beta H} = \prod_{i=1}^N \mathcal{T}_{s_i s_{i+1}}, \quad \mathcal{T}_{ss'} = \exp \left[K s s' + \frac{\beta h}{2} (s + s') \right].$$

$\mathcal{T}$ is a 2×2 matrix, the *transfer matrix*:

$$\mathcal{T} = \begin{pmatrix} e^{K+\beta h} & e^{-K} \\ e^{-K} & e^{K-\beta h} \end{pmatrix}.$$

The sum over all configurations is a product of matrices:

$$Z = \sum_{s_1, \dots, s_N} \mathcal{T}_{s_1 s_2} \mathcal{T}_{s_2 s_3} \cdots \mathcal{T}_{s_N s_1} = \text{Tr } \mathcal{T}^N = \lambda_+^N + \lambda_-^N, \quad (8)$$

where $\lambda_{\pm}$ are the eigenvalues of $\mathcal{T}$:

$$\lambda_{\pm} = e^K \cosh \beta h \pm \sqrt{e^{2K} \sinh^2 \beta h + e^{-2K}}. \quad (9)$$

Since $\lambda_+ > |\lambda_-|$, for large N only the largest eigenvalue survives: $Z \simeq \lambda_+^N$ and $f = -k_B T \ln \lambda_+$. Equation (8) is exact for any N , which gives exact results for finite chains as well; the program uses it to check the simulations.

Key result

The Ising chain in a field has the magnetization and zero-field susceptibility

$$m = \frac{\sinh \beta h}{\sqrt{\sinh^2 \beta h + e^{-4K}}}, \quad \chi|_{h=0} = \frac{e^{2J/k_B T}}{k_B T}. \quad (10)$$

At $h = 0$, $m = 0$ at every temperature $T > 0$: the chain has no spontaneous magnetization.

Compared with free spins, the interaction makes the chain much more responsive: the susceptibility is enhanced by the factor $e^{2J/k_B T}$, which grows without bound as $T \rightarrow 0$ (Figure 1). But it only diverges at $T = 0$.

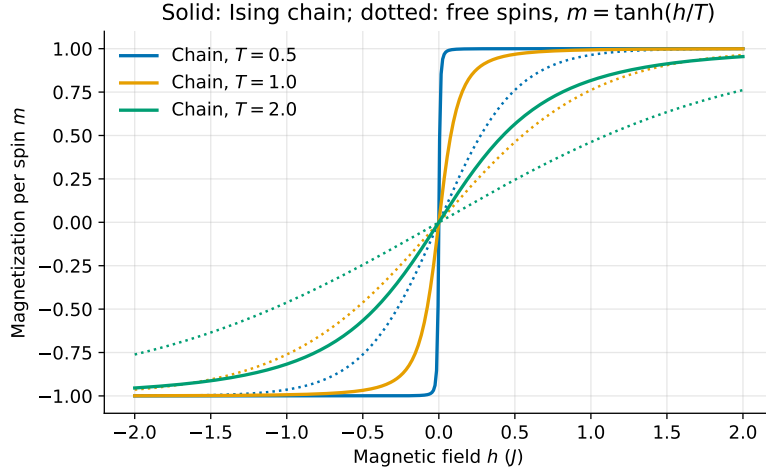


Figure 1: Magnetization of the Ising chain in a field, Eq. (10) (solid lines), compared with independent spins, $m = \tanh(h/T)$ (dotted). At low temperature the chain responds to a very small field, because the spins are aligned over long distances, but $m = 0$ at $h = 0$.

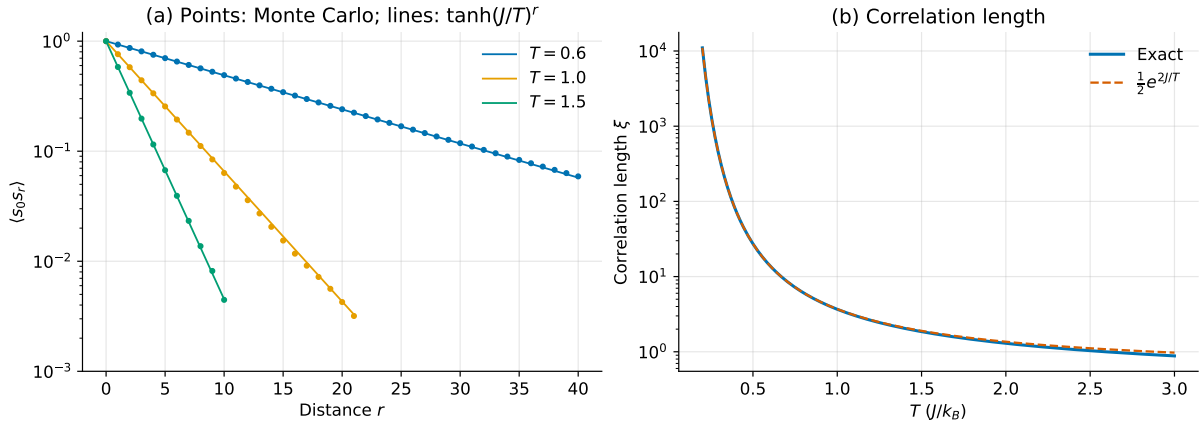


Figure 2: (a) Correlation function of the chain, measured in simulations of 400 spins (points), compared with the exact result $(\tanh K)^r$ (lines). (b) The correlation length (11) and its low-temperature approximation.

3.3 Correlations and the correlation length

How far does the order extend? The *spin-spin correlation function* $G(r) = \langle s_i s_{i+r} \rangle$ measures how much the spin at site $i + r$ “knows” about the spin at site i . With bond variables, $s_i s_{i+r} = \tau_i \tau_{i+1} \cdots \tau_{i+r-1}$, and since the τ are independent with $\langle \tau \rangle = \tanh K$,

$$G(r) = (\tanh K)^r = e^{-r/\xi}, \quad \xi = -\frac{1}{\ln \tanh K} \underset{T \rightarrow 0}{\simeq} \frac{1}{2} e^{2J/k_B T}. \quad (11)$$

The correlations decay exponentially over the *correlation length* ξ , the typical size of the ordered domains. As $T \rightarrow 0$, ξ grows exponentially, but it is finite at every $T > 0$ (Figure 2).

The susceptibility is related to the correlations. From Eq. (4) at $h = 0$,

$$k_B T \chi = \frac{1}{N} \sum_{i,j} \langle s_i s_j \rangle = \sum_r G(r) = 1 + 2 \sum_{r=1}^{\infty} (\tanh K)^r = \frac{1 + \tanh K}{1 - \tanh K} = e^{2K},$$

which is Eq. (10) again. This relation is general: *a large susceptibility means long-range correlations*. At a critical point, where χ diverges, the correlation length diverges too.

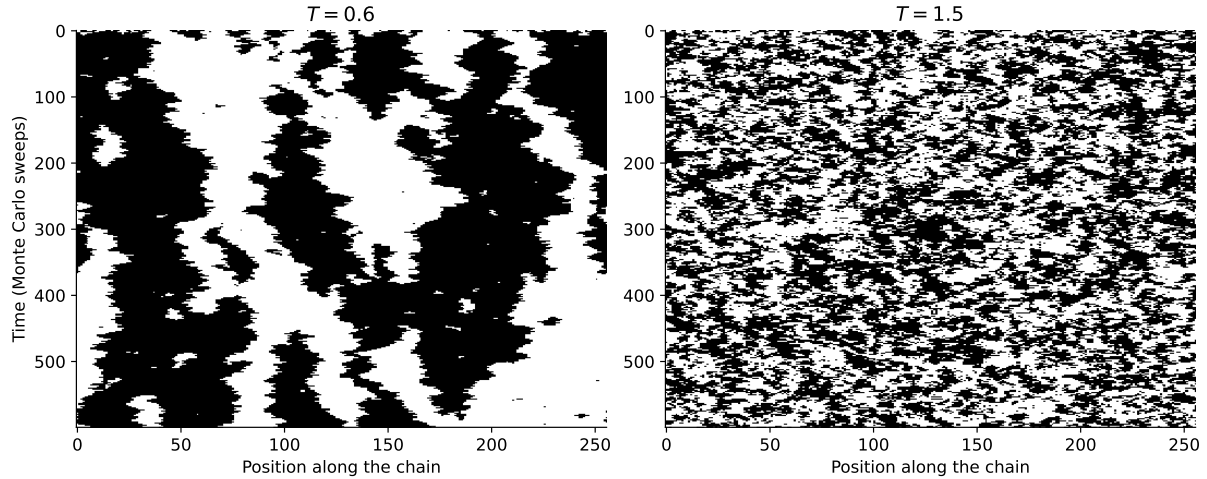


Figure 3: Space-time picture of a chain of 256 spins in a Metropolis simulation, starting from random spins (white: up, black: down). Each row is the chain after one sweep. At $T = 0.6$ ($\xi \approx 14$), domains with a mean length of about 29 spins form and slowly diffuse; at $T = 1.5$ ($\xi \approx 1.9$), the mean domain length is about 5.

3.4 Why there is no phase transition in one dimension

A simple argument, due to Landau, explains the absence of order. Start from the fully ordered chain of N spins and flip all the spins to the right of some bond. This creates a *domain wall*, which costs an energy $2J$. The wall can be placed on any of the $N - 1$ bonds, which gives an entropy $k_B \ln(N - 1) \simeq k_B \ln N$. The free energy changes by

$$\Delta F = 2J - k_B T \ln N,$$

which is negative for any $T > 0$ when N is large enough. Domain walls are therefore always created, and they destroy the long-range order. Their density is $1/(1 + e^{2K}) \simeq e^{-2J/k_B T}$ per bond, about one wall every two correlation lengths. Figure 3 shows the domains appearing, moving and annihilating in a simulation. In two dimensions, the argument fails, as we will see: a wall is a line whose energy grows with its length.

4 Mean-field theory

4.1 The Weiss molecular field

In most cases, the model cannot be solved exactly. The simplest approximation, due to Pierre Weiss (1907), replaces the neighbours of each spin by their average value. Spin i then feels an effective field

$$h_{\text{eff}} = h + J \sum_{j \text{ neighbour of } i} \langle s_j \rangle = h + qJm,$$

and behaves like an independent spin in this field (Section 2.4). Its average must equal m , which gives the *self-consistency equation*

$$m = \tanh(\beta(qJm + h)). \quad (12)$$

At $h = 0$, $m = 0$ is always a solution. The graphical solution (Figure 4a) shows that two other solutions $\pm m_0$ appear when the slope of $\tanh(\beta qJm)$ at the origin exceeds 1, that is, below the mean-field critical temperature

$$k_B T_c^{\text{MF}} = qJ. \quad (13)$$

4.2 Critical behaviour

Near T_c^{MF} , m is small and we can expand $\tanh x \simeq x - x^3/3$. With $h = 0$, Eq. (12) gives

$$m \simeq \sqrt{3} \left(1 - \frac{T}{T_c^{\text{MF}}}\right)^{1/2}.$$

Above T_c^{MF} , a small field gives $m \simeq \beta(qJm + h)$, so that $\chi = m/h = 1/[k_B(T - T_c^{\text{MF}})]$: the Curie–Weiss law. At $T = T_c^{\text{MF}}$, $m \simeq (3h/qJ)^{1/3}$. These power laws define *critical exponents* (Section 5.4); mean-field theory predicts the exponents $\beta = 1/2$, $\gamma = 1$ and $\delta = 3$. (By tradition, the exponent β has the same name as the inverse temperature $\beta = 1/k_B T$; the context always makes clear which one is meant.)

4.3 Landau theory

The same results follow from the mean-field free energy per spin,

$$f(m) = \frac{1}{2}qJm^2 - k_B T \ln[2 \cosh(\beta(qJm + h))],$$

whose minimum is given by Eq. (12). Expanding at $h = 0$ in powers of m ,

$$f(m) \simeq -k_B T \ln 2 + \frac{qJ}{2} \left(1 - \frac{T_c^{\text{MF}}}{T}\right) m^2 + \frac{k_B (T_c^{\text{MF}})^4}{12 T^3} m^4. \quad (14)$$

Above T_c^{MF} the coefficient of m^2 is positive and the minimum is at $m = 0$. Below T_c^{MF} it is negative and f has two symmetric minima at $\pm m_0$: this is spontaneous symmetry breaking. Lev Landau showed that the form (14) follows from symmetry alone, whatever the microscopic details, which explains why mean-field theory gives the same exponents for all systems.

4.4 Successes and failures

Mean-field theory describes the existence of the transition and the shape of the phase diagram qualitatively, but it neglects fluctuations, which are essential near T_c :

- in one dimension it predicts a transition at $k_B T_c = 2J$, whereas there is none;
- in two dimensions it predicts $k_B T_c = 4J$, whereas the exact value is $2.269 J$ (Figure 4b);
- its critical exponents are wrong in two and three dimensions.

Above four dimensions, the *upper critical dimension* of the Ising model, the mean-field critical exponents become exact, although the critical temperature itself remains below qJ/k_B . Mean-field theory is fully exact when each spin interacts equally with all the others.

5 The two-dimensional Ising model

5.1 The Peierls argument: order at low temperature

In two dimensions, a domain of reversed spins is bounded by a closed wall. A wall of length ℓ (in lattice spacings) breaks ℓ bonds, and costs an energy $2J\ell$. The number of closed walls of length ℓ through a given bond grows roughly like 3^ℓ , since at each step the wall can continue in 3 directions. The free energy of such walls is therefore

$$\Delta F(\ell) \approx \ell (2J - k_B T \ln 3),$$

which is positive and grows with ℓ when $k_B T < 2J/\ln 3 = 1.82 J$. Large reversed domains are then suppressed, and the system keeps a spontaneous magnetization. Peierls proposed this

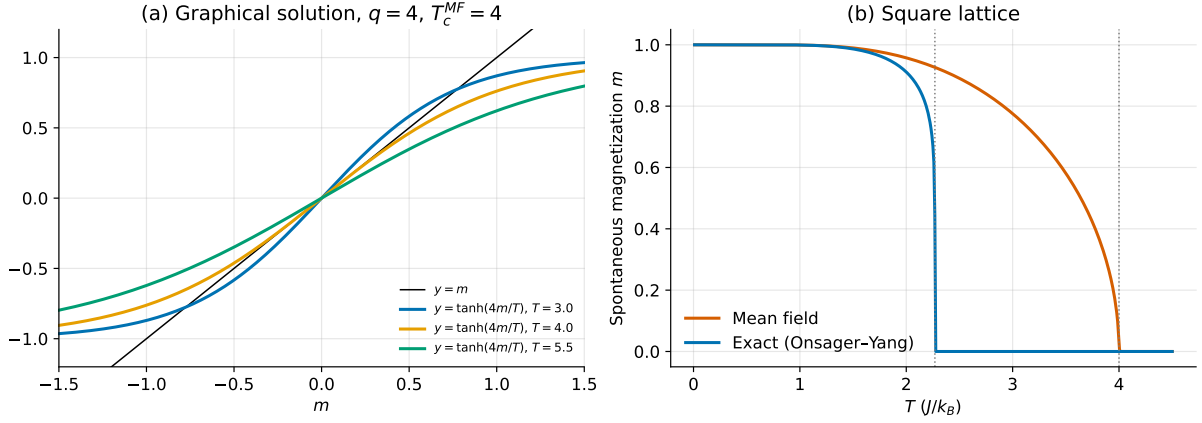


Figure 4: (a) Graphical solution of the mean-field equation (12) on the square lattice ($q = 4$) at $h = 0$: non-zero solutions exist for $T < 4$. (b) Mean-field spontaneous magnetization compared with the exact result of Onsager and Yang, Eq. (20).

argument in 1936 [3]; a fully rigorous version, which proves that the 2D model is ordered at low temperature, was completed in the 1960s by Griffiths and Dobrushin. The estimate $2J/\ln 3$ is crude: the exact value $k_B T_c = 2.269 J$ is higher, since counting all walls as independent overestimates their entropy. The contrast with one dimension is essential: there, a wall is a point whose energy does not grow with the size of the domain.

5.2 Kramers–Wannier duality: the critical temperature

The exact critical temperature can be found from a hidden symmetry. There are two ways to write the partition function.

High-temperature expansion. Since $s_i s_j = \pm 1$, $e^{K s_i s_j} = \cosh K (1 + u s_i s_j)$ with $u = \tanh K$. Expanding the product over the $2N$ bonds of the square lattice gives a sum over sets of bonds, each weighted by $u^{\text{number of bonds}}$. The sum over each spin vanishes unless the spin appears an even number of times, so only sets of bonds forming *closed graphs* survive:

$$Z = 2^N (\cosh K)^{2N} \sum_{\text{closed graphs } G} u^{|G|}.$$

Low-temperature expansion. A configuration is described by its domain walls, which are closed graphs on the *dual lattice*, whose sites are the centres of the squares. Each wall bond costs $2J$, so

$$Z = 2 e^{2NK} \sum_{\text{closed graphs } G^*} e^{-2K|G^*|}.$$

The dual of the square lattice is again a square lattice, so both sums are the same function $\Phi(x) = \sum_G x^{|G|}$, evaluated at $x = \tanh K$ and at $x = e^{-2K}$. The partition function at a high temperature (small K) is therefore related to that at a low temperature K^* such that

$$\tanh K = e^{-2K^*} \iff \sinh 2K \sinh 2K^* = 1.$$

If there is a single phase transition, it must occur at the self-dual point $K = K^*$:

$$\sinh \frac{2J}{k_B T_c} = 1, \quad k_B T_c = \frac{2J}{\ln(1 + \sqrt{2})} = 2.269185 \dots J. \quad (15)$$

5.3 Onsager's solution

In 1944, Onsager computed the free energy of the infinite square lattice at $h = 0$ exactly [5]. With $K = \beta J$ and $\kappa = 2 \sinh 2K / \cosh^2 2K$, his result can be written

$$-\beta f = \ln(2 \cosh 2K) + \frac{1}{\pi} \int_0^{\pi/2} \ln \frac{1 + \sqrt{1 - \kappa^2 \sin^2 \phi}}{2} d\phi. \quad (16)$$

Derivatives give the energy and the specific heat, in terms of the complete elliptic integrals $K_1(\kappa) = \int_0^{\pi/2} d\phi / \sqrt{1 - \kappa^2 \sin^2 \phi}$ and $E_1(\kappa) = \int_0^{\pi/2} \sqrt{1 - \kappa^2 \sin^2 \phi} d\phi$:

$$e = -J \coth 2K \left[1 + \frac{2}{\pi} \left(2 \tanh^2 2K - 1 \right) K_1(\kappa) \right], \quad (17)$$

$$\frac{c}{k_B} = \frac{4}{\pi} (K \coth 2K)^2 \left\{ K_1 - E_1 - \left(1 - \tanh^2 2K \right) \left[\frac{\pi}{2} + \left(2 \tanh^2 2K - 1 \right) K_1 \right] \right\}. \quad (18)$$

At T_c , $\kappa = 1$ and K_1 diverges logarithmically. The energy is continuous, $e(T_c) = -\sqrt{2} J$, but the specific heat diverges logarithmically:

$$\frac{c}{k_B} \simeq -\frac{8K_c^2}{\pi} \ln \left| 1 - \frac{T}{T_c} \right| = -0.4945 \ln \left| 1 - \frac{T}{T_c} \right|. \quad (19)$$

The spontaneous magnetization, announced by Onsager in 1948–49 and derived by Yang in 1952 [7], is

Key result

$$m_0 = \left(1 - \frac{1}{\sinh^4 2K} \right)^{1/8} \quad (T < T_c), \quad m_0 = 0 \quad (T > T_c). \quad (20)$$

Near T_c , $m_0 \simeq 1.222 (1 - T/T_c)^{1/8}$: the magnetization vanishes with the exponent $\beta = 1/8$.

These results are the black curves in Figure 7. The program computes them in the module `exact.py`, and the tests check that Eqs. (17) and (18) are indeed the derivatives of Eq. (16).

5.4 Critical exponents, scaling and universality

Near the critical point, the thermodynamic quantities behave as power laws of the reduced temperature $t = (T - T_c)/T_c$, which define the *critical exponents*:

$$\begin{aligned} c &\sim |t|^{-\alpha}, & m_0 &\sim (-t)^\beta, & \chi &\sim |t|^{-\gamma}, & \xi &\sim |t|^{-\nu}, \\ m &\sim h^{1/\delta} \text{ at } t = 0, & G(r) &\sim r^{-(d-2+\eta)} \text{ at } t = 0. \end{aligned}$$

Table 1 compares their values for the 2D and 3D Ising models and for mean-field theory.

	α	β	γ	δ	ν	η
2D Ising (exact)	0 (log)	1/8	7/4	15	1	1/4
3D Ising [15]	0.110	0.326	1.237	4.79	0.630	0.036
Mean field	0 (jump)	1/2	1	3	1/2	0

Table 1: Critical exponents of the Ising model. The 3D values are numerical estimates.

The exponents are not independent. The *scaling hypothesis*, according to which the only relevant length near T_c is ξ , implies the relations

$$\alpha + 2\beta + \gamma = 2, \quad \gamma = \beta(\delta - 1), \quad \gamma = \nu(2 - \eta), \quad d\nu = 2 - \alpha,$$

which the exact 2D exponents satisfy (Exercise 8).

The exponents do not depend on the lattice (square, triangular, honeycomb), nor on the details of the interactions, nor even on the nature of the system: they depend only on the dimension and on the symmetry of the order parameter. This *universality* explains why the liquid–gas critical points of real fluids have the exponents of the 3D Ising model. It is explained by the *renormalization group* of Kadanoff and Wilson: near T_c , the physics at scales much larger than the lattice spacing forgets the microscopic details.

5.5 Finite lattices

A finite lattice has no true phase transition: ξ cannot exceed the linear size L . The singularities are rounded: the specific heat and the susceptibility have finite peaks, whose heights grow with L and whose positions tend to T_c . Kaufman’s solution [6] gives the partition function of a finite periodic $L \times L$ lattice exactly. With $K = \beta J$,

$$Z = \frac{1}{2}(2 \sinh 2K)^{L^2/2} \sum_{i=1}^4 Z_i, \quad Z_{1,2} = \prod_{k=0}^{L-1} 2 \left\{ \frac{\cosh}{\sinh} \right\} \left(\frac{L}{2} \gamma_{2k+1} \right), \quad Z_{3,4} = \prod_{k=0}^{L-1} 2 \left\{ \frac{\cosh}{\sinh} \right\} \left(\frac{L}{2} \gamma_{2k} \right),$$

where $\cosh \gamma_k = \cosh 2K \coth 2K - \cos(\pi k/L)$ for $k > 0$ and $\gamma_0 = 2K + \ln \tanh K$. The program uses this formula (checked against the direct enumeration of all 2^{16} states of a 4×4 lattice) to give exact finite-size curves for the energy and specific heat: the dashed lines of Figure 7. How the peaks grow with L is itself governed by the critical exponents; this *finite-size scaling* is the main tool to extract exponents from simulations (Section 8.4).

6 Monte Carlo simulation

6.1 Importance sampling

Since we cannot sum over all configurations, we sample them. Choosing configurations uniformly at random would be useless: almost all of them have $\mathcal{M} \approx 0$ and an energy close to zero, and they have an exponentially small weight at low temperature. The idea of *importance sampling* is to generate configurations $\sigma_1, \sigma_2, \dots, \sigma_n$ directly with the Boltzmann probability (2). Averages are then simple arithmetic means,

$$\langle A \rangle \approx \frac{1}{n} \sum_{k=1}^n A(\sigma_k),$$

with a statistical error that decreases like $1/\sqrt{n}$.

6.2 Markov chains and detailed balance

We cannot draw configurations from $P(\sigma)$ directly, since we do not know Z . Instead we build a *Markov chain*: starting from any configuration, we go from σ to σ' with a transition probability $W(\sigma \rightarrow \sigma')$. If

1. every configuration can be reached from every other one, and the chain does not cycle periodically (*ergodicity*), and
2. the transition probabilities satisfy *detailed balance*,

$$P(\sigma) W(\sigma \rightarrow \sigma') = P(\sigma') W(\sigma' \rightarrow \sigma), \quad (21)$$

then after many steps the chain visits the configurations with the probability $P(\sigma)$, whatever the starting point. Detailed balance ensures that P is stationary: the probability flow from σ to σ' equals the reverse flow. For the Boltzmann distribution it only involves the ratio $P(\sigma')/P(\sigma) = e^{-\beta \Delta E}$, where $\Delta E = H(\sigma') - H(\sigma)$, so Z is not needed.

6.3 The Metropolis algorithm

The Metropolis algorithm [10] proposes to flip one spin at a time. Flipping spin i changes the energy by

$$\Delta E = 2s_i \left(J \sum_{j \text{ nb of } i} s_j + h \right), \quad (22)$$

which only involves its neighbours. The flip is accepted with probability

$$A = \min \left(1, e^{-\beta \Delta E} \right). \quad (23)$$

A flip that lowers the energy is always accepted; one that raises it is accepted with the Boltzmann factor. The ratio of the forward and backward probabilities is $e^{-\beta \Delta E}$, so detailed balance holds (Exercise 9).

Algorithm 1 One Metropolis sweep

```

1: for  $N$  times do
2:   choose a site  $i$  at random
3:    $\Delta E \leftarrow 2s_i \left( J \sum_{j \in \text{nb}(i)} s_j + h \right)$ 
4:   if  $\Delta E \leq 0$  or  $\text{random}() < e^{-\beta \Delta E}$  then
5:      $s_i \leftarrow -s_i$  ▷ accept the flip
6:   end if
7: end for

```

One *sweep*, N attempted flips, gives every spin on average one chance to flip; time in Monte Carlo simulations is measured in sweeps. On the square lattice at $h = 0$, ΔE only takes the values $-8J, -4J, 0, 4J, 8J$.

6.4 The heat-bath algorithm on a checkerboard

In the heat-bath algorithm, spin i is not flipped but drawn anew from its conditional probability given its neighbours: it is set to $+1$ with probability

$$p_+ = \frac{1}{1 + e^{-2\beta(J \sum_j s_j + h)}},$$

whatever its current value. This also satisfies detailed balance. The lattice can be coloured like a chessboard: the neighbours of a black site are all white. All the black spins can therefore be updated at the same time, then all the white spins. This *checkerboard* update is well suited to vectorized computing; the program implements it with NumPy.

A pitfall

One might be tempted to apply the Metropolis rule (23) to all the black spins at the same time. Each sub-lattice update does leave the Boltzmann distribution invariant, but the resulting Markov chain is *not ergodic*. A spin with $\Delta E = 0$ is flipped with probability exactly 1, and the dynamics becomes partly deterministic. For a ring of 6 spins, the transition matrix of this update has three eigenvalues of modulus 1 instead of one: the chain is trapped in subsets of configurations. The simulations then converge to wrong averages. For a chain of 8 spins at $T = 1$, starting with all spins up, the energy per spin comes out as -0.880 instead of the exact -0.818 ; the wrong value even depends on the starting configuration. The heat-bath rule, whose probabilities are never exactly 0 or 1, does not have this problem. Checking against exact results is the only way to catch such errors.

6.5 Equilibration and measurements

A simulation starts from an arbitrary configuration, either random (“hot start”) or fully ordered (“cold start”). The first sweeps are discarded until the chain has forgotten its starting point (*thermalization*). The energy and magnetization are then measured after every sweep. Figure 12a shows typical time series of $|m|$.

6.6 Statistical errors

Successive configurations of a Markov chain are correlated, so the n measurements are not independent. The *autocorrelation function* of an observable A ,

$$C_A(t) = \frac{\langle A_k A_{k+t} \rangle - \langle A \rangle^2}{\langle A^2 \rangle - \langle A \rangle^2},$$

decays over a time of order the *integrated autocorrelation time* $\tau = \frac{1}{2} + \sum_{t \geq 1} C_A(t)$. The effective number of independent measurements is $n/(2\tau)$, and the error on $\langle A \rangle$ is $\sqrt{2\tau}$ times larger than the naive $\sigma_A/\sqrt{n}$.

In practice the program uses *binning*: the series is cut into 32 blocks, much longer than τ , whose averages are independent. For quantities that are non-linear functions of averages, such as $c \propto \langle e^2 \rangle - \langle e \rangle^2$, the error is estimated with the *jackknife*: the quantity is recomputed 32 times, each time leaving out one block, and the spread of these values gives the error.

6.7 Critical slowing down and the Wolff algorithm

Near T_c , the system is made of correlated domains of all sizes up to ξ . Single-spin flips change them only at their edges, and the autocorrelation time grows like $\tau \sim \xi^z$, where z is the *dynamic exponent*. At T_c on a finite lattice, $\xi \sim L$ and $\tau \sim L^z$. For local algorithms, $z \approx 2.17$ [14]: simulations of large lattices near T_c become very slow. This is *critical slowing down*.

Cluster algorithms solve this problem by flipping whole domains at once. In the Wolff algorithm [12], based on ideas of Swendsen and Wang [11]:

1. choose a random seed spin;
2. add each neighbour that is parallel to a spin of the cluster with probability $p = 1 - e^{-2\beta J}$, and repeat from the newly added spins until the cluster stops growing;
3. flip the whole cluster.

The move is always accepted. Detailed balance holds because the probability of building the same cluster in the forward and backward moves differs only by the bonds on its boundary that were not added. Their ratio, $(1-p)^{n_1}/(1-p)^{n_2} = e^{-2\beta J(n_1-n_2)}$, where n_1 and n_2 are the numbers of parallel boundary bonds before and after the flip, is exactly $e^{-\beta \Delta E}$ (Exercise 11). Near T_c the clusters have the size of the correlated domains, and the autocorrelation time barely grows with L (Section 8.6). At high temperature, clusters are single spins; at low temperature, they fill the lattice.

A pitfall

Clusters have random sizes, so a Wolff “sweep” must be defined with care. It is tempting to flip clusters until N spins have been flipped, then measure. But the moment of the measurement then depends on the size of the last cluster, and the measured configurations are biased. We made this mistake while writing the code, and the energy of a 4×4 lattice came out more than 100 standard deviations away from the exact value. The correct procedure is to flip a *fixed* number of clusters between measurements. The program determines this number during thermalization from the average cluster size.

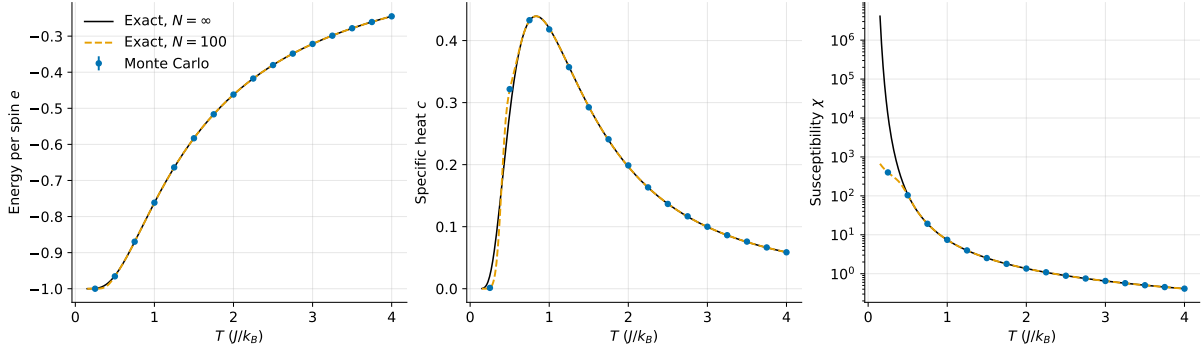


Figure 5: Energy, specific heat and susceptibility per spin of the Ising chain. Points: Monte Carlo simulations of 100 spins with the Wolff algorithm (10^5 sweeps per temperature). Lines: exact transfer-matrix results for 100 spins and for the infinite chain.

6.8 Checking the code against exact results

Table 2 compares the three algorithms with the exact results of a 4×4 lattice, obtained by enumerating its $2^{16} = 65\,536$ configurations. The results agree with the exact values: 34 of the 36 deviations are below two standard deviations, and the largest is 2.6 standard deviations. Repeating the runs with 16 different random seeds confirms that the deviations are statistical fluctuations: they average to zero, with a spread of one error bar to within 10–20%, which is the accuracy of the error estimates themselves. The test suite of the program performs such checks automatically, also for rings of spins, where the transfer matrix gives exact results.

T		e	$\langle m \rangle$	c	χ
1.500	Exact	-1.95064	0.98617	0.20035	0.02949
	Metropolis	-1.9507(6)	0.98620(19)	0.200(3)	0.0294(9)
	Heat bath	-1.9509(4)	0.98628(15)	0.199(2)	0.0291(7)
	Wolff	-1.9510(3)	0.98629(9)	0.1990(15)	0.0290(5)
2.269	Exact	-1.56562	0.84386	0.78327	0.34732
	Metropolis	-1.5697(15)	0.8454(7)	0.778(3)	0.345(2)
	Heat bath	-1.5684(19)	0.8450(9)	0.780(4)	0.345(3)
	Wolff	-1.5663(11)	0.8441(5)	0.780(2)	0.3462(14)
3.000	Exact	-1.01707	0.60129	0.60313	0.47700
	Metropolis	-1.016(2)	0.6004(11)	0.6042(13)	0.4788(9)
	Heat bath	-1.0184(19)	0.6024(10)	0.6043(11)	0.4763(10)
	Wolff	-1.0166(12)	0.6013(6)	0.6023(11)	0.4764(8)

Table 2: Monte Carlo results for a 4×4 lattice (4×10^5 sweeps) compared with exact enumeration. The number in parentheses is the statistical error on the last digits.

7 Results in one dimension

Figure 5 shows the energy, specific heat and susceptibility of a chain of 100 spins, simulated with the Wolff algorithm, compared with the exact transfer-matrix results. The agreement is perfect within the error bars. The finite size only matters at low temperature, where the correlation length becomes comparable to the length of the chain: the susceptibility then saturates at $\chi \simeq N/T$, the value for a fully aligned chain, instead of growing like $e^{2J/T}/T$. The correlation function (Figure 2) decays exponentially at every temperature, as expected in the absence of a phase transition.

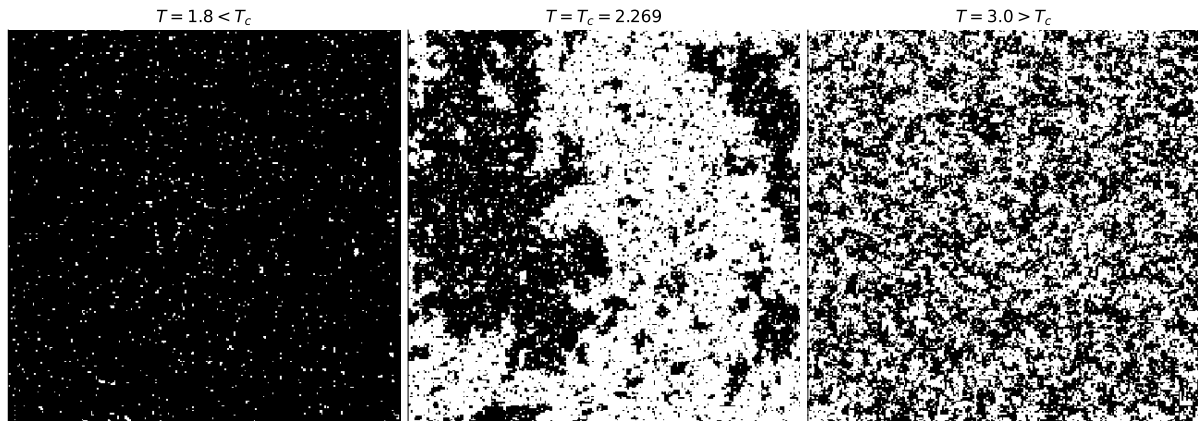


Figure 6: Equilibrium configurations of a 256×256 lattice (white: up, black: down), below, at and above the critical temperature.

Try it with the code

`python3 ising_1D.py` reproduces these results (options 1 and 3), shows the magnetization in a field (option 2) and the space-time picture of Figure 3 (option 4). Try increasing the number of spins at low temperature: how does the susceptibility change?

8 Results in two dimensions

8.1 Configurations

Figure 6 shows typical configurations of a 256×256 lattice. Below T_c , the lattice is almost fully magnetized, with small islands of reversed spins. Above T_c , the spins form small domains of typical size ξ , with no net magnetization. At T_c , domains of all sizes coexist, nested within each other: the configuration looks the same at all scales. This *scale invariance* is the hallmark of a critical point.

8.2 Thermodynamics

Figure 7 shows the energy, magnetization, specific heat and susceptibility for $L = 8$ to 64 . The simulated energy and specific heat agree with Kaufman's exact results for the same lattices, and approach Onsager's infinite-lattice curves as L grows. Over all sizes and temperatures, the deviations from Kaufman's values have a spread of about one error bar, the largest being about three error bars, as expected for more than a hundred comparisons. The magnetization $\langle |m| \rangle$ follows Yang's formula below T_c , but does not vanish above it: in a finite lattice, $\langle |m| \rangle \sim 1/\sqrt{N}$ at high temperature. The specific heat and the susceptibility have peaks that grow with L and approach T_c .

8.3 Correlations at the critical point

Away from T_c , correlations decay exponentially. At T_c , the correlation length is infinite and the correlation function decays as a power law, $G(r) \sim r^{-1/4}$ ($\eta = 1/4$). Figure 8 shows both behaviours in a 256×256 lattice.

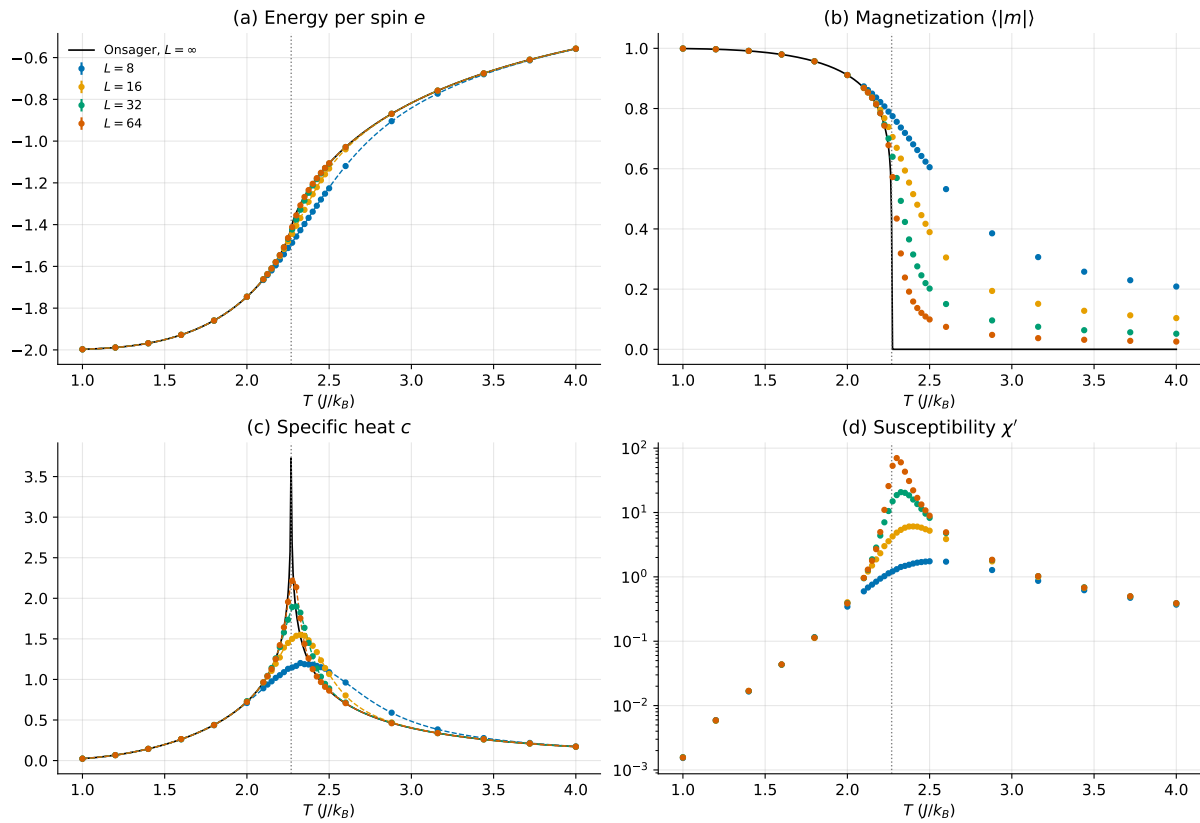


Figure 7: Thermodynamics of the 2D Ising model. Points: Wolff simulations (10^5 sweeps per temperature) for $L = 8, 16, 32$ and 64 . Dashed lines: Kaufman's exact results for the same lattices. Black lines: Onsager's and Yang's exact results for the infinite lattice. The dotted line marks T_c .

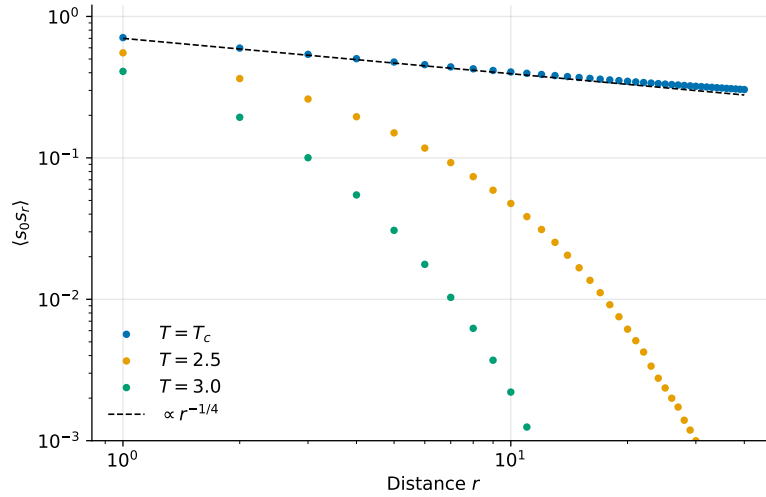


Figure 8: Spin-spin correlation function of a 256×256 lattice (log-log scale). At T_c it decays as the power law $r^{-1/4}$; above T_c it decays exponentially.

8.4 Finite-size scaling and the Binder cumulant

How do the finite-lattice quantities depend on L near T_c ? The *finite-size scaling* hypothesis states that the only relevant lengths are $\xi \sim |t|^{-\nu}$ and L . For the susceptibility, for example,

$$\chi'(T, L) = L^{\gamma/\nu} \tilde{\chi}(L^{1/\nu} t), \quad (24)$$

where $\tilde{\chi}$ is a universal function. Consequences:

- at T_c , $\langle |m| \rangle \sim L^{-\beta/\nu} = L^{-1/8}$ and $\chi' \sim L^{\gamma/\nu} = L^{7/4}$;
- the maximum of χ' grows like $L^{7/4}$, and its position approaches T_c like $L^{-1/\nu} = L^{-1}$;
- plotting $\chi' L^{-7/4}$ against $L t$ makes the curves of all sizes collapse onto one.

Figure 9 shows this *data collapse*. Figure 10 shows the power laws: the fits give $\beta/\nu = 0.125$ and $\gamma/\nu = 1.770$, close to the exact $1/8 = 0.125$ and $7/4 = 1.75$. The maxima of χ' are obtained from dedicated simulations around each peak. The small excess of γ/ν comes from corrections to scaling at small L : fitting only $L = 32, 64$ and 128 gives 1.759 .

A convenient way to locate T_c is the *Binder cumulant* [13]

$$U_L = 1 - \frac{\langle m^4 \rangle}{3 \langle m^2 \rangle^2}. \quad (25)$$

Deep in the ordered phase, $m \simeq \pm m_0$ and $U_L \rightarrow 2/3$. In the disordered phase, m has Gaussian fluctuations, for which $\langle m^4 \rangle = 3 \langle m^2 \rangle^2$, and $U_L \rightarrow 0$. At T_c , U_L takes a universal value independent of L , so the curves for different sizes cross at T_c (Figure 9a). Here the curves for $L = 64$ and $L = 128$ cross at $T = 2.266$ and $U = 0.615$, very close to $T_c = 2.269$.

Try it with the code

Option 5 of `python3 ising_2D.py` computes the Binder cumulant for $L = 8$ to 64 . With these data, estimate T_c from the crossing points, and γ/ν from the maxima of χ .

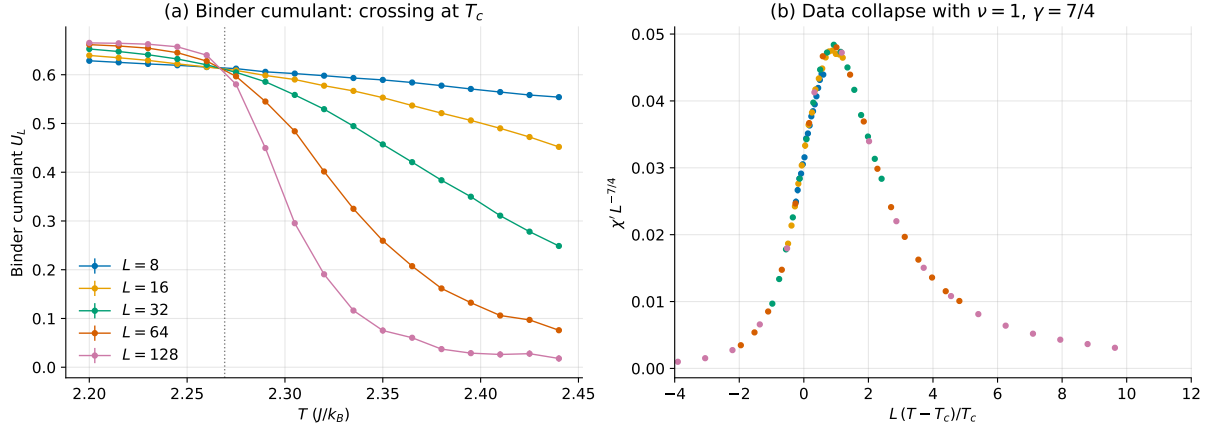


Figure 9: (a) Binder cumulant for $L = 8$ to 128 : the curves cross at the critical temperature (dotted line). (b) Finite-size scaling of the susceptibility, Eq. (24), with the exact exponents $\nu = 1$ and $\gamma = 7/4$: the data for all sizes collapse onto a single curve.

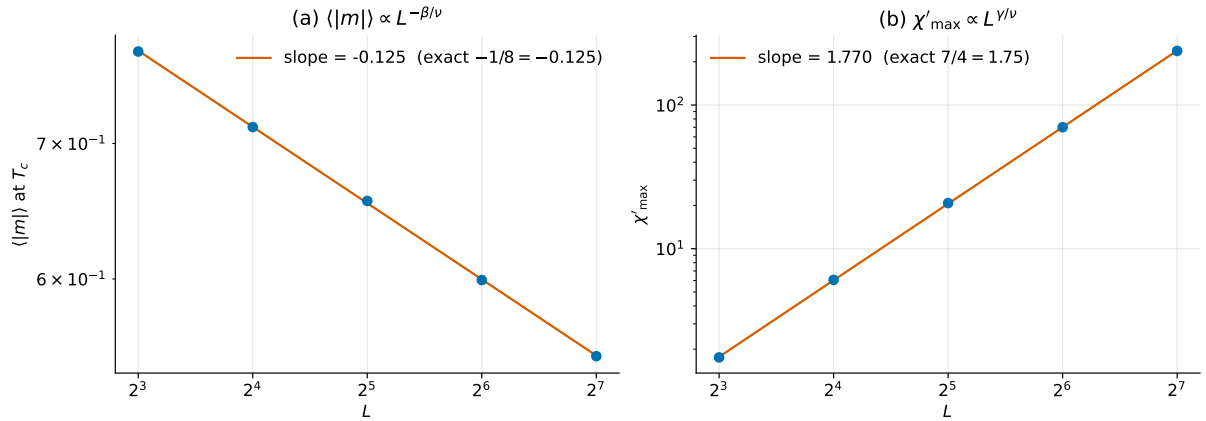


Figure 10: (a) Magnetization at T_c and (b) maximum of the susceptibility as functions of L (log-log scale). The slopes give $-\beta/\nu$ and γ/ν .

8.5 Hysteresis

Below T_c , the system has two ordered states, $m \approx \pm m_0$. If the field is swept from positive to negative values, the lattice stays magnetized up while the field is slightly negative. The up state is still stable against small fluctuations, and a reversed domain must be large enough to grow. The magnetization flips only at a finite *coercive field*. Sweeping back gives the symmetric branch, and the magnetization traces a *hysteresis loop* (Figure 11). The loop shrinks as T increases and disappears above T_c , where m is a single-valued function of h : at $T = 3$ the two branches coincide within the statistical noise. Hysteresis is a non-equilibrium effect, and the width of the loop depends on how fast the field is swept. At $T = 2$, sweeping four times more slowly (160 instead of 40 sweeps per field value) reduces the coercive field from about 0.20 to about 0.12: the slower the sweep, the more time a reversed domain has to nucleate and grow. In true equilibrium, the magnetization of an infinite system would jump from $-m_0$ to $+m_0$ at $h = 0$.

8.6 Critical slowing down

Figure 12 compares the Metropolis and Wolff algorithms at T_c . With Metropolis, $|m|$ wanders slowly and its autocorrelation time grows like $L^{2.09}$, close to the known dynamic exponent $z \approx 2.17$. With Wolff, the autocorrelation time, measured in sweeps, stays of order one and grows only like $L^{0.40}$. For $L = 64$, the autocorrelation time is 718 sweeps with Metropolis and

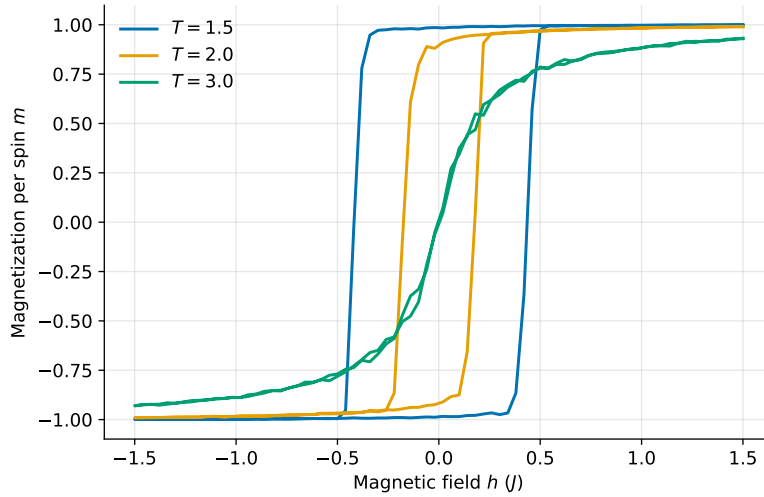


Figure 11: Magnetization of a 64×64 lattice while the field is swept from $+1.5$ to -1.5 and back, in steps of 0.04 . At each field value, 20 Metropolis sweeps let the lattice relax and the next 20 are used to measure m .

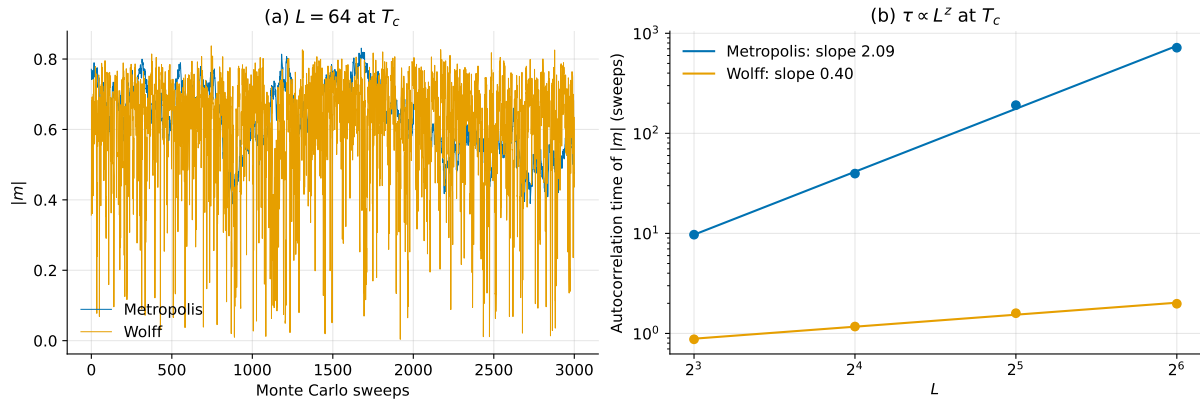


Figure 12: (a) Time series of $|m|$ for a 64×64 lattice at T_c with the Metropolis and Wolff algorithms. (b) Integrated autocorrelation time of $|m|$ at T_c as a function of L .

2.0 sweeps with Wolff, about 362 times shorter. A Wolff sweep costs 1.1 times more computer time, so Wolff is still about 316 times more efficient. This is why all the scans near T_c in these notes use it.

9 Beyond

- **Three dimensions.** The 3D Ising model has never been solved exactly. On the simple cubic lattice, the most precise simulations give $k_B T_c = 4.5115 J$ [16], and the exponents of Table 1. The methods of these notes (Wolff clusters, Binder cumulant, finite-size scaling) are exactly those used to obtain these numbers.
- **Other spin models.** Spins with more than two states give the Potts model, and spins that can point in any direction of a plane or of space give the XY and Heisenberg models, each with its own universality class.
- **Quantum Ising model.** Spins in a transverse magnetic field undergo a phase transition at zero temperature, driven by quantum fluctuations. The quantum chain is equivalent to the classical 2D model.

- **Renormalization group.** Grouping spins into blocks and looking at how the effective couplings change with the block size explains universality and gives a way to compute the exponents.

For more, see the textbooks of Yeomans [18] and Kardar [19] on phase transitions, Baxter [20] on exact solutions, and Newman and Barkema [21] and Landau and Binder [22] on Monte Carlo methods.

10 Using the code

The programs are in the repository <https://github.com/daoudizakaria/Ising-Model>. They require Python 3 with `numpy`, `scipy` and `matplotlib`; installing `numba` compiles the Metropolis and Wolff algorithms and makes them much faster.

ising_1D.py Interactive program for the chain: thermodynamics, magnetization in a field, correlation function, space-time picture.

ising_2D.py Interactive program for the square lattice: snapshots, animation, thermodynamics compared with Onsager and Kaufman, hysteresis, Binder cumulant, critical slowing down.

ising/ The package: the lattices and algorithms (`lattice.py`), exact results (`exact.py`), statistical analysis (`analysis.py`) and simulation drivers (`simulate.py`).

make_figures.py Produces all the figures of these notes.

tests/ Tests of the code against exact results (`python3 -m unittest discover tests`).

For example, a simulation of a 32×32 lattice at $T = 2$ takes three lines:

```
from ising import run, exact
r = run(dim=2, size=32, T=2.0, n_therm=2000, n_meas=20000, algorithm="wolff")
print(r["abs_m"], exact.onsager_magnetization(2.0)) # (value, error), exact
```

11 Exercises

1. **Counting.** How many configurations does a 10×10 Ising lattice have? How long would it take to enumerate them at 10^9 configurations per second? Compare with the age of the Universe, 1.4×10^{10} years.
2. **Fluctuations.** Derive Eqs. (3) and (4) from Z .
3. **Independent spins.** For $J = 0$, compute Z , m , the energy and the specific heat per spin. Show that $\chi = 1/k_B T$ at $h = 0$.
4. **Specific heat of the chain.** Show that the specific heat (7) is maximum when $K \tanh K = 1$, and find the corresponding temperature.
5. **Transfer matrix.** Check the eigenvalues $\lambda_{\pm}$ of $\mathcal{T}$, and derive the magnetization (10) from $m = \partial \ln \lambda_+ / \partial(\beta h)$. What are its limits for $T \rightarrow 0$ and $T \rightarrow \infty$?
6. **Mean field.** (a) Derive $m \simeq \sqrt{3}(1 - T/T_c)^{1/2}$ and the Curie–Weiss law from Eq. (12). (b) Show that the mean-field specific heat jumps from $\frac{3}{2}k_B$ to 0 at T_c .
7. **Duality.** Solve $\tanh K = e^{-2K}$ and check Eq. (15).
8. **Scaling relations.** Check that the exact 2D exponents and the mean-field exponents (in $d = 4$) satisfy the four scaling relations of Section 5.4.

- 9. Detailed balance.** Show that the Metropolis acceptance (23) satisfies detailed balance (21). Do the same for the heat-bath probability.
- 10. Acceptance rates.** On the square lattice at $h = 0$, list the possible values of ΔE and the corresponding Metropolis acceptance probabilities at $T = 1$ and $T = 3$.
- 11. Wolff algorithm.** Consider a cluster C with n_1 parallel bonds on its boundary before the flip and n_2 after. Show that the energy changes by $\Delta E = 2J(n_1 - n_2)$, and that the ratio of the probabilities to build C in the forward and backward moves is $(1 - p)^{n_1 - n_2}$. Deduce that the choice $p = 1 - e^{-2\beta J}$ satisfies detailed balance with an acceptance of 1.
- 12. Lattice gas.** With $s_i = 2n_i - 1$, show that the Ising Hamiltonian becomes $H = -4J \sum_{\langle ij \rangle} n_i n_j - \mu \sum_i n_i + \text{const}$, and express μ in terms of h and J (on the square lattice).
- 13. With the code.** (a) Using `ising_2D.py`, option 3, measure the height of the specific heat peak for $L = 8, 16, 32$ and 64 , and show that it grows like $\ln L$. (b) Using option 5, estimate T_c from the crossings of the Binder cumulant.

Answers

1. $2^{100} \approx 1.3 \times 10^{30}$; $1.3 \times 10^{21} \text{ s} = 4 \times 10^{13} \text{ years}$, three thousand times the age of the Universe.
2. $\partial_\beta \ln Z = -\langle H \rangle$ and $\partial_\beta^2 \ln Z = \langle H^2 \rangle - \langle H \rangle^2$; with $\partial_T = -k_B \beta^2 \partial_\beta$, $C = k_B \beta^2 (\langle H^2 \rangle - \langle H \rangle^2)$. Similarly with derivatives with respect to βh .
3. $Z = (2 \cosh \beta h)^N$, $m = \tanh \beta h$, $e = -h \tanh \beta h$, $c = k_B (\beta h)^2 / \cosh^2 \beta h$.
4. $dc/dK = 0$ gives $K \tanh K = 1$, $K = 1.200$, $k_B T = 0.834 J$, $c = 0.439 k_B$.
5. $\det(\mathcal{T} - \lambda) = 0$ gives $\lambda^2 - 2\lambda e^K \cosh \beta h + 2 \sinh 2K = 0$. For $T \rightarrow 0$, $m \rightarrow \text{sign}(h)$; for $T \rightarrow \infty$, $m \simeq \tanh(\beta h)$, free spins.
6. (a) See Section 4. (b) Below T_c , $e = -\frac{qJ}{2} m^2 \simeq -\frac{3qJ}{2} (1 - T/T_c)$, so $c = \frac{3qJ}{2T_c} = \frac{3}{2} k_B$ just below T_c , and $c = 0$ above.
7. With $x = e^{-2K}$, $(1-x)/(1+x) = x$ gives $x^2 + 2x - 1 = 0$, $x = \sqrt{2} - 1$, $K_c = \frac{1}{2} \ln(1 + \sqrt{2}) = 0.4407$.
8. 2D: $0 + \frac{1}{4} + \frac{7}{4} = 2$; $\frac{1}{8} \times 14 = \frac{7}{4}$; $1 \times (2 - \frac{1}{4}) = \frac{7}{4}$; $2 \times 1 = 2 - 0$. Mean field in $d = 4$: $0 + 1 + 1 = 2$; $\frac{1}{2} \times 2 = 1$; $\frac{1}{2} \times 2 = 1$; $4 \times \frac{1}{2} = 2$.
9. If $\Delta E > 0$, $W(\sigma \rightarrow \sigma') = e^{-\beta \Delta E}$ and $W(\sigma' \rightarrow \sigma) = 1$, whose ratio is $P(\sigma')/P(\sigma)$. Heat bath: the ratio of p_+ and p_- is $e^{2\beta(J \sum s_j + h)}$, the ratio of the Boltzmann weights.
10. $\Delta E = -8, -4, 0$: always accepted. $\Delta E = 4$: $e^{-4} = 0.018$ at $T = 1$, $e^{-4/3} = 0.26$ at $T = 3$. $\Delta E = 8$: 3.4×10^{-4} and 0.069 .
11. Bonds inside the cluster are unchanged; boundary bonds switch between parallel and antiparallel. The boundary bonds that are parallel must all have been rejected: $(1 - p)^{n_1}$ forward and $(1 - p)^{n_2}$ backward. With $1 - p = e^{-2\beta J}$ the ratio is $e^{-2\beta J(n_1 - n_2)} = e^{-\beta \Delta E}$.
12. $H = -4J \sum n_i n_j - (2h - 8J) \sum n_i + \text{const}$ (each site has 4 neighbours), so $\mu = 2h - 8J$.

13. (a) Kaufman's exact values of the maximum of c are 1.19, 1.55, 1.90 and 2.25 for $L = 8, 16, 32$ and 64 : each doubling of L adds about 0.35, close to $(8K_c^2/\pi)\ln 2 = 0.343$, so $c_{\max} \simeq 0.49 \ln L + \text{const.}$ The simulated points agree with these exact curves (Figure 7c). The positions of the maxima, 2.362, 2.318, 2.294 and 2.282, approach T_c like $1/L$, as expected for $\nu = 1$. (b) The crossings are within about 0.01 of $T_c = 2.269$.

References

- [1] W. Lenz, "Beitrag zum Verständnis der magnetischen Erscheinungen in festen Körpern," *Physikalische Zeitschrift* **21**, 613 (1920).
- [2] E. Ising, "Beitrag zur Theorie des Ferromagnetismus," *Zeitschrift für Physik* **31**, 253 (1925).
- [3] R. Peierls, "On Ising's model of ferromagnetism," *Proceedings of the Cambridge Philosophical Society* **32**, 477 (1936).
- [4] H. A. Kramers and G. H. Wannier, "Statistics of the two-dimensional ferromagnet. Part I," *Physical Review* **60**, 252 (1941).
- [5] L. Onsager, "Crystal statistics. I. A two-dimensional model with an order-disorder transition," *Physical Review* **65**, 117 (1944).
- [6] B. Kaufman, "Crystal statistics. II. Partition function evaluated by spinor analysis," *Physical Review* **76**, 1232 (1949).
- [7] C. N. Yang, "The spontaneous magnetization of a two-dimensional Ising model," *Physical Review* **85**, 808 (1952).
- [8] T. D. Lee and C. N. Yang, "Statistical theory of equations of state and phase transitions. II. Lattice gas and Ising model," *Physical Review* **87**, 410 (1952).
- [9] S. G. Brush, "History of the Lenz–Ising model," *Reviews of Modern Physics* **39**, 883 (1967).
- [10] N. Metropolis, A. W. Rosenbluth, M. N. Rosenbluth, A. H. Teller, and E. Teller, "Equation of state calculations by fast computing machines," *Journal of Chemical Physics* **21**, 1087 (1953).
- [11] R. H. Swendsen and J.-S. Wang, "Nonuniversal critical dynamics in Monte Carlo simulations," *Physical Review Letters* **58**, 86 (1987).
- [12] U. Wolff, "Collective Monte Carlo updating for spin systems," *Physical Review Letters* **62**, 361 (1989).
- [13] K. Binder, "Finite size scaling analysis of Ising model block distribution functions," *Zeitschrift für Physik B* **43**, 119 (1981).
- [14] M. P. Nightingale and H. W. J. Blöte, "Dynamic exponent of the two-dimensional Ising model and Monte Carlo computation of the subdominant eigenvalue of the stochastic matrix," *Physical Review Letters* **76**, 4548 (1996).
- [15] A. Pelissetto and E. Vicari, "Critical phenomena and renormalization-group theory," *Physics Reports* **368**, 549 (2002).
- [16] A. M. Ferrenberg, J. Xu, and D. P. Landau, "Pushing the limits of Monte Carlo simulations for the three-dimensional Ising model," *Physical Review E* **97**, 043301 (2018).
- [17] J. J. Hopfield, "Neural networks and physical systems with emergent collective computational abilities," *Proceedings of the National Academy of Sciences* **79**, 2554 (1982).
- [18] J. M. Yeomans, *Statistical Mechanics of Phase Transitions*, Oxford University Press (1992).
- [19] M. Kardar, *Statistical Physics of Fields*, Cambridge University Press (2007).

- [20] R. J. Baxter, *Exactly Solved Models in Statistical Mechanics*, Academic Press (1982).
- [21] M. E. J. Newman and G. T. Barkema, *Monte Carlo Methods in Statistical Physics*, Oxford University Press (1999).
- [22] D. P. Landau and K. Binder, *A Guide to Monte Carlo Simulations in Statistical Physics*, 5th ed., Cambridge University Press (2021).